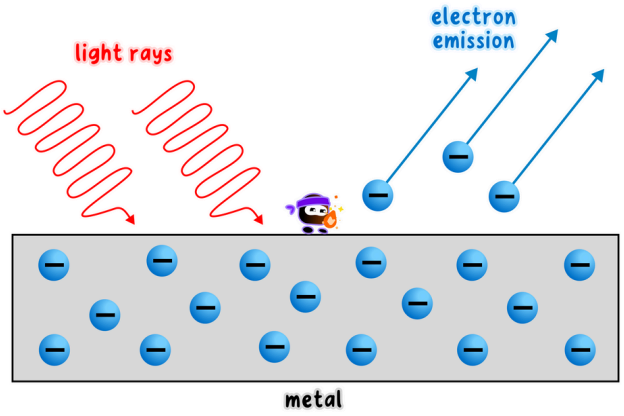
IB PHYSICS E.2 QUANTUM PHYSICS (HL ONLY)

Photoelectric effect

- Light incident on a metal surface → electrons ejected if **frequency ≥ threshold**.
- **Below threshold frequency** → no emission regardless of intensity.
- **Above threshold frequency** →
  - Kinetic energy of electrons ∝ frequency (not intensity).
  - Emission is **instantaneous** (no time lag).
  - Higher intensity → more electrons emitted (higher current), but **same max KE**.
- **Wave model failed**:
  - Couldn't explain threshold frequency.
  - Predicted intensity should increase KE - not observed.
  - Predicted time lag at low intensity - not observed.



Einstein's photon explanation & work function

- **Photon energy:** E = hf (h = Planck's constant).
- Photon hits electron → energy used to overcome **work function Φ** (binding energy).
- Remainder → **max kinetic energy** of emitted electron:

$$E_{\max} = hf - \Phi$$

- **Threshold frequency**:

$$f_t = \frac{\Phi}{h}$$

- Below f_t → no emission, even at high intensity.
- **Stopping potential (V_s):** Voltage needed to stop electrons →

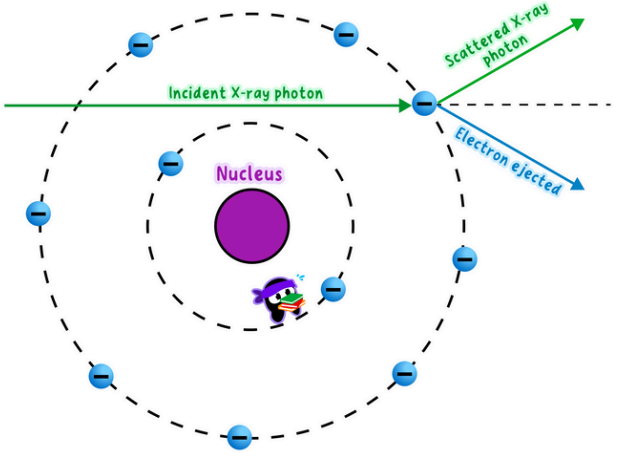
$$eV_s = E_{\max}$$

Compton scattering

- X-rays scatter off electrons → **wavelength increases** after collision.
- Explained by treating photons as particles with **energy & momentum**.
- Wavelength shift depends on scattering angle θ:

$$\Delta\lambda = \lambda_f - \lambda_i = \frac{h}{m_e c}(1 - \cos\theta)$$

- **Largest shift** at θ = 180° (backscatter).
- Demonstrates **momentum transfer** between photon and electron.



Threshold frequency and instantaneous emission

- **Threshold frequency f_t**: minimum frequency to eject electrons.
- At $f = f_t$ → emitted electrons have **zero kinetic energy**.
- Instantaneous emission → each photon interacts with a single electron.

De Broglie hypothesis

- Proposed by Louis de Broglie (1923).
- Any particle with momentum p has wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where p = mv for non-relativistic speeds.

- Predicts **wave behavior** (diffraction, interference) for particles.
- Bridge between classical (particle) and quantum (wave) descriptions.



Note!

Larger momentum = shorter wavelength = less noticeable wave behavior.

Particle diffraction

- Davisson-Germer experiment (1927): electrons accelerated at Ni crystal.
- Produced **diffraction pattern** similar to X-rays.
- Observed **spacing** matched predicted de Broglie wavelength.
- Proved **electrons** behave as **waves** under suitable conditions.

Wave-particle duality

- Light → **wave phenomena** (diffraction, interference) + **particle phenomena** (photoelectric effect, Compton).
- Matter → **particle behavior** (mass, momentum) + **wave behavior** (diffraction).
- Quantum physics treats light and matter as having **dual nature**, depending on the phenomenon.

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